

INSTALLING rBROWSER2

Installation, starting it and first setup

Windows - Linux - macOS - Android

Version 2.0.6b. Installer, pip by hand, venv,
Docker and Termux, the start options,
the first setup, how to update it
and the most common errors on every system.

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BEFORE YOU START

This guide is for rBrowser2 version 2.0.6b.

rBrowser2 is a browser for the Reticulum network: it opens NomadNet pages, and it also has the LXMF messenger, the RRC chat, the BBS client and the RMAP map built in. It is a Python program that runs on your computer (or on a Raspberry Pi, or on Android with Termux) and you use it from your web browser: Chrome, Firefox, Edge, whichever you like.

You need:

- Python 3.9 or newer, with pip;
- Reticulum (RNS): rBrowser installs it itself, together with the other packages it needs;
- git (recommended, not required): it lets you update rBrowser with one click from its page.

If you have never installed Python, get it from python.org first (on Windows tick "Add python.exe to PATH" in the installer). This guide assumes that "python3 --version" (on Windows "python --version") answers.

The commands below are written with "python3" and "pip3": use the ones that work on YOUR computer.

0. WHICH WAY TO CHOOSE

There are several ways to install rBrowser2. Pick ONE:

a) INSTALLER (recommended, above all the first time): a script that does everything by itself. Sections 2 (Windows) and 3 (Linux, Mac).

b) BY HAND, with pip: if Reticulum is already set up and you know what you are doing. Section 4.

c) BY HAND IN A VENV: like b), but the packages stay in a folder of their own, without touching the system Python. Section 5.

d) DOCKER: rBrowser in a container (Linux, Raspberry Pi). Section 6.

e) ANDROID with Termux. Section 7.

In every case the first step is downloading rBrowser2 (section 1).

1. DOWNLOADING rBROWSER2

1.1 With git (recommended)

Open the terminal in the folder where you want it and:

```
git clone https://github.com/fr33n0w/rBrowser2.git
cd rBrowser2
```

With git you can update rBrowser with one click from its page (section 10).

If git is missing:

```
Windows:  https://git-scm.com/download/win  (install it, then
          open a NEW terminal)
Linux:    sudo apt install git
macOS:    git --version  (if it is missing, the Mac offers to
          install the "Command Line Tools": accept)
```

1.2 Without git (zip)

- Go to https://github.com/fr33n0w/rBrowser2
- Green "Code" button > "Download ZIP" (or the release zip, on the "Releases" page).
- Extract the zip into a folder of your choice.
- Open the terminal INSIDE that folder (the one with rBrowser.py in it).

With the zip there is no update from the page: you update by hand (section 10).

On Windows, to open the terminal in the folder: open the folder in File Explorer, click the address bar at the top, type cmd and press Enter.

2. WINDOWS: THE INSTALLER (install.bat)

2.1 Running it

In the rBrowser2 folder double-click:

```
install.bat
```

or run it from the Command Prompt, in the folder:

```
install.bat
```

2.2 What it does

- looks for Python 3.9 or newer (tries "python", then "py -3");
- installs the packages in requirements.txt (Reticulum, LXMF, Flask and the others);
- checks the Reticulum configuration: if you have none, it writes a basic one with the rmap.world backbone; if you have one, it asks whether to add rmap.world (or switch it on, if it is there but off). The other interfaces stay as they were, and the old file is saved as config.before-rbrowser;
- asks whether to also install the APP VERSION (default: no): rBrowser in a window of its own, with an icon in the tray, instead of a tab of your web browser (section 8.2);
- at the end it tells you how to start rBrowser.

Everything stays in the rBrowser2 folder. You can run it again whenever you like, it does no harm.

2.3 Starting rBrowser

In the Command Prompt, in the rBrowser2 folder:

```
python rBrowser.py
```

(or py rBrowser.py if you use "py")

Then open your browser at: http://localhost:5000

With the app version: python rBrowser.py --app (section 8.2).

To keep it running, the terminal window must stay open (you can minimise it).

2.4 Common errors on Windows

- **"Python 3.9 or newer is needed"**

  Python is missing or not in the PATH: reinstall it from python.org and tick "Add python.exe to PATH".

- **"pip install failed"**

  Read the error above it. Often upgrading pip and running it again is enough:

  ```
  python -m pip install --upgrade pip
  ```

- **Windows asks for Firewall permission at the first start**

  Allow it, at least for private networks, if you want to open rBrowser from your phone or other PCs at home too. If you only use it from this PC, you can also deny it.

3. LINUX, RASPBERRY PI AND macOS: THE INSTALLER (install.sh)

3.1 Running it

In the terminal, in the rBrowser2 folder:

```
bash install.sh
```

3.2 What it does

- looks for Python 3.9 or newer and pip (on Debian/Ubuntu/Pi OS, if pip is missing, it offers to install it with apt);
- installs the packages in requirements.txt, choosing WHERE:
  - if a venv is ACTIVE, it installs there, no questions asked;
  - if rBrowser's venv already exists (.venv, from an earlier install), it uses that one;
  - on "protected" systems (Debian 12+, Ubuntu 23.04+, Raspberry Pi OS Bookworm and Trixie) it ASKS YOU:
    1) a venv in the rBrowser folder (.venv): it touches nothing outside the folder. You then start it with .venv/bin/python rBrowser.py
    2) your user (~/.local) with --break-system-packages, as the Reticulum manual does: rnsd and the other Reticulum commands shared with your other programs.

  If you are not sure: 1 is the cleanest, 2 is the handiest if you also use NomadNet, Sideband or other Reticulum programs on the same computer.
- checks the Reticulum configuration as on Windows (a basic config with rmap.world if you have

none; if you have one, it offers to add it or switch it on; it saves the old file as config.before-rbrowser). On Linux it uses BackboneInterface (the fastest TCP), on macOS TCPClientInterface;

- puts rnsd, rnstatus and the other Reticulum commands in the PATH, if they are not there (typical on the Raspberry Pi). Open a new terminal for it to apply there too;
- ONLY WITH A SCREEN (macOS, or a Linux desktop; not Termux, not a Pi reached over ssh): it asks whether to also install the APP VERSION, rBrowser in a window of its own (default: no). On Debian/Ubuntu/Pi OS it adds GTK and WebKit with apt (sudo password). Section 8.2;
- ONLY ON LINUX with systemd: it asks whether to start rBrowser as a system service at every boot (default answer: no). It asks for the sudo password. Without the service you can switch the start at boot on later from the page (section 9.7).

3.3 Starting rBrowser

Installed for your user (or with a venv of yours active):

```
python3 rBrowser.py
```

Installed in rBrowser's venv (choice 1):

```
.venv/bin/python rBrowser.py
```

With the system service: it has already started, and it starts again by itself at every boot. Useful commands:

```
systemctl status rbrowser
journalctl -u rbrowser -f
sudo systemctl stop rbrowser
sudo systemctl disable rbrowser
```

(the last one removes the start at boot)

Then open your browser at: http://localhost:5000

3.4 Common errors on Linux and macOS

- **"Python 3.9 or newer is needed"**

  Debian, Ubuntu, Pi OS: sudo apt install python3 python3-pip

  macOS: Python from https://www.python.org

- **"Could not make the venv: install python3-venv"**

  On Debian/Ubuntu/Pi OS the venv module is a separate package:

  ```
  sudo apt install python3-venv
  ```

  then run bash install.sh again.

- **After the install "rnsd: command not found"**

  Open a new terminal (the installer added the folder to the PATH, but only terminals opened afterwards see it).

- **Raspberry Pi Zero (512 MB of RAM) freezing during the install**

  Install with nothing heavy running (stop any Docker containers): with everything at once the memory runs out.

- **Raspberry Pi OS: "ERROR: pip's dependency resolver ... types-flask-migrate"**

  It is about a system package, not rBrowser's: ignore it.

4. BY HAND, WITH pip (Reticulum already set up)

If you already have Reticulum and it works, you do not need the installer. In the rBrowser2 folder:

4.1 Windows

```
pip install -r requirements.txt
python rBrowser.py
```

(if "pip" is missing: python -m pip install -r requirements.txt)

4.2 macOS and "unprotected" Linux (Arch, Fedora, older Mint...)

```
pip3 install -r requirements.txt
python3 rBrowser.py
```

4.3 Debian 12+, Ubuntu 23.04+, Raspberry Pi OS (Bookworm, Trixie)

Here pip refuses with "externally-managed-environment". Install for your user, as the Reticulum manual does:

```
pip3 install --user --break-system-packages -r requirements.txt
python3 rBrowser.py
```

The commands pip installs (rnsd, rnstatus...) end up in ~/.local/bin: if the terminal does not find them, open a new one, or:

```
echo 'export PATH="$HOME/.local/bin:$PATH"' >> ~/.bashrc
source ~/.bashrc
```

You cannot or do not want to use --break-system-packages (e.g. a work PC)? Use a venv: section 5.

5. BY HAND, IN A VENV

A venv is a folder with a Python of its own: rBrowser's packages go in there and the system Python stays untouched. It works on every system. In the rBrowser2 folder:

5.1 Linux and macOS

```
python3 -m venv .venv
.venv/bin/pip install -r requirements.txt
.venv/bin/python rBrowser.py
```

On Debian/Ubuntu/Pi OS, if the first command fails:

```
sudo apt install python3-venv
```

The next times only the last command is needed:

```
.venv/bin/python rBrowser.py
```

Or you can "activate" the venv: after this command "python3" and "pip3" are the venv's, until "deactivate":

```
source .venv/bin/activate
```

5.2 Windows

```
python -m venv .venv
.venv\Scripts\pip install -r requirements.txt
.venv\Scripts\python rBrowser.py
```

The next times only the last command is needed.

5.3 What to know about the venv

- rnsd, rnstatus and the other Reticulum commands are INSIDE the venv (.venv/bin/ on Linux and Mac, .venv\Scripts\ on Windows):

```
.venv/bin/rnstatus
```

- If you use other Reticulum programs (NomadNet, Sideband...) installed outside the venv, that is fine: rBrowser connects to the Reticulum already running, if there is one (section 8.3).
- The installer (bash install.sh) recognises the .venv venv and reuses it.

6. DOCKER (Linux, Raspberry Pi)

6.1 The quickest way

In the rBrowser2 folder:

```
bash docker.sh
```

It installs Docker if missing (with apt), builds the image and starts rBrowser. To stop it:

```
bash docker.sh stop
```

Then open: http://localhost:5000

6.2 By hand (Docker already installed)

```
sudo usermod -aG docker $USER
```

(then LOG OUT and log in again, for the group to apply)

```
docker compose up -d
docker compose logs -f rbrowser
```

Other commands:

```
docker compose down          (stop)
docker compose restart       (restart)
docker compose build --no-cache rbrowser   (rebuild)
```

With the "docker-compose" package the commands are written with a dash: docker-compose up -d

6.3 What to know about Docker

- Identities, settings and saved pages live in three Docker volumes: they survive rebuilds and "docker compose down". Only "docker compose down -v" deletes them.
- The Reticulum configuration inside the container is the "config" file in the rBrowser2 folder. To use yours, put it in its place before building, or mount it in docker-compose.yaml.
- Port and address are changed in docker-compose.yaml, not from the page; Restart and Shut down on the page are disabled: use docker compose restart / stop.
- Raspberry Pi Zero 2 W: the first build takes about 9 minutes.

7. ANDROID WITH TERMUX

7.1 Installing Termux

Get Termux from F-Droid (https://f-droid.org/packages/com.termux/): the Play Store version is old. Open it once and wait for the setup to finish.

7.2 Downloading rBrowser2 and running the installer

```
pkg update && pkg upgrade
pkg install git
git clone https://github.com/fr33n0w/rBrowser2.git
cd rBrowser2
bash install.sh
```

The installer adds the Termux packages (Python, psutil, the prebuilt cryptography and the build tools, should pip need them), the Python packages and the Reticulum configuration with rmap.world.

7.3 Starting rBrowser

```
python3 rBrowser.py --host 127.0.0.1
```

Then open in the phone's browser: http://127.0.0.1:5000

With --host 127.0.0.1 rBrowser opens only from the phone itself. Without --host it also opens from the other devices on your home network.

To stop Android putting Termux to sleep:

```
termux-wake-lock
```

(and take Termux out of battery optimisation in the Android settings). To stop rBrowser: Ctrl+C, then termux-wake-unlock.

7.4 Common errors on Termux

- **pip stops with "Failed to determine Android API level"**

  ```
  export ANDROID_API_LEVEL=$(getprop ro.build.version.sdk)
  ```

  then run bash install.sh again.

- **"Permission denied" / "Address already in use" when Reticulum starts**

  An old Reticulum is still running. Stop it and try again:

  ```
  pkill -f reticulum
  ```

8. STARTING rBROWSER: ALL THE OPTIONS

8.1 The command

In the rBrowser2 folder (Windows: python instead of python3, venv: .venv/bin/python):

```
python3 rBrowser.py
```

Options (all optional):

```
--host 127.0.0.1      only from this computer
--host 0.0.0.0        also from the other devices at home
--port 5080           another port (default: 5000)
--log full            the terminal shows everything
--log status          only connection, identity and errors
--open LINK           opens a nomadnetwork:// or rrc:// link in
                      the running rBrowser (starts it if needed)
--register-links      nomadnetwork://, rrc:// and rbrowser://
                      links open with rBrowser, then it exits
                      (Windows and Linux)
--unregister-links    removes that association, then it exits
--app                 opens rBrowser in a window of its own
                      (app version, section 8.2)
--info                shows the state of the rBrowser already
                      running here, then follows its log
                      (with --port N for one on another port)
```

Without --host and --port, what you saved on the page applies (Settings > System Setup > Web server); the first time 0.0.0.0 and port 5000. Options written in the command apply only to that run.

8.2 The app version (--app)

Installed with the installer ("app version" question) or by hand:

```
pip3 install -r requirements-app.txt
```

(on Linux also: sudo apt install python3-gi gir1.2-webkit2-4.1) then:

```
python3 rBrowser.py --app
```

rBrowser opens in a window of its own, with an icon in the tray, instead of a tab of the web browser. Closing the window stops rBrowser. Phones and other PCs still open it in their browser, as usual. Windows and Linux desktops; macOS not tested yet. Settings > System Setup > Shortcuts puts an rBrowser2 icon on the desktop and in the app menu, which opens the app version.

8.3 Reticulum: do I need to start rnsd first?

No. When it starts, rBrowser connects to the Reticulum already running on the computer (rnsd, or NomadNet, Sideband, MeshChat...) if there is one; otherwise it starts its own. If you use several Reticulum programs on the same computer, the tidiest way is to keep rnsd running and let them all connect to it.

8.4 Opening rBrowser

On this computer: http://localhost:5000

From a phone or another PC at home: http://PC-ADDRESS:5000

The PC's address on the home network (like 192.168.1.20):

```
Windows:  ipconfig        ("IPv4 Address" line)
Linux:    hostname -I
macOS:    System Settings > Wi-Fi > Details
```

Or switch on the network name rbrowser.local (section 9.6) and open http://rbrowser.local:5000 without remembering the address. "rBrowser2 running on" in Settings > System Setup and in About lists all the addresses it can be opened at.

The same address opens from several devices at once: it is always the same rBrowser, with the same identities and the same messages. One instance per folder: do not start rBrowser twice from the same folder.

8.5 Stopping it

Ctrl+C in the terminal, or the "Shut down" button at the bottom of Settings. "Restart" restarts it.

8.6 Common errors at start

- **The port is already in use ("Address already in use", "only one usage of each socket address")**

  An rBrowser (or another program) is already running on that port. Use that one, or start with another port:

  ```
  python3 rBrowser.py --port 5080
  ```

- **"Attempt to reinitialise Reticulum, when it was already running"**

  Another program started its own Reticulum with the same settings. Close it, or start rnsd and let them all connect to it.

- **The page does not open from the phone**

  - Was rBrowser started with --host 127.0.0.1? Then it opens only from the computer: drop --host (or use 0.0.0.0).
  - Are the phone and the computer on the same Wi-Fi network?
  - Windows: the Firewall must allow Python on private networks.

- **"ModuleNotFoundError: No module named ..."**

  The packages are installed in another Python (or in the venv and you start it outside the venv, or the other way round). Start rBrowser with the same Python you ran pip install with.

- **"rnsd runs under another user" / "digest sent was rejected"**

  rnsd was started by another user (a system service, say) and answers only programs with its key: rBrowser stays on without Reticulum. As the user of rnsd, this prints the key:

  ```
  python3 -c "import RNS, os
  p = os.path.expanduser('~/.reticulum/storage/transport_identity')
  k = RNS.Identity.from_file(p).get_private_key()
  print(RNS.Identity.full_hash(k).hex())"
  ```

  Then in ~/.reticulum/config of rBrowser's user, under [reticulum], add the line:

  ```
  rpc_key = THE-KEY
  ```

  and restart rBrowser. The key gives control of rnsd: keep it private.

9. FIRST SETUP

Everything is set from the page: the gear icon (Settings). Settings are saved in the rBrowser2 folder (settings/), and survive updates.

9.1 At the first start

Only the NomadNet browser is on: it starts listening to announces straight away, and nodes appear in the list on the left as they announce (on a quiet network this can take a few minutes). You can also paste a node's address in the bar at the top, without waiting. LXMF Messenger, RRC chat, BBS client and RMAP are OFF. This is on purpose: each one is set up and personalised at the first start (your name, your nick and the rest), then you switch it on.

The status bar at the top says whether Reticulum is connected, how many announces and nodes have arrived, CPU and RAM, the battery (where there is one), the rBrowser version and the time.

9.2 Access password (Settings > System Setup > Access password)

Off at first. rBrowser starts on 0.0.0.0, that is reachable from your whole home network: anyone on your network can open it and use your identities. If the network is not only yours, SWITCH ON the password. From the computer itself it does not ask for it, unless you choose so in the same section. It is set, and later changed, from its own window (Change password).

9.3 Identities (Settings > Identity)

rBrowser creates its "Default" identity by itself. Here you can create more, import them, export them (make a backup!) and choose which one to identify with to nodes. Nodes see your identity only if you identify (Identify, or Auto-identify for the nodes you choose).

9.4 Switching the clients on

Each one has the "Enable" switch at the top of its Settings:

- LXMF Messenger: first set your name ("Name in announces"), then switch it on. It is Reticulum's messenger, compatible with Sideband, MeshChat, NomadNet.
- RRC Client: the room chat. BEFORE switching it on choose your nick: at first the "Default nick" is "Anonymous Peer rB2", and it is on. Write yours there. If you switch the nick off, the others only see the code (hash) of your identity.
- BBS Client: the Reticulum bulletin boards.
- RMAP: the map of Reticulum nodes (rmap.world).

The icons of the clients that are on appear next to the page tabs (Settings > System Setup > Apps in the tab bar).

9.5 Is Reticulum connected? The servers

If the status bar does not say it is connected, or no node arrives, check Reticulum in the terminal:

```
rnstatus
```

There must be at least one interface towards the network (e.g. rmap.world) with "Status: Up". Tip: in the Reticulum config put SEVERAL servers (at least 3), so you stay connected if one goes down. The list is on https://rmap.world and in rBrowser's RMAP client. How to add them is in the Reticulum manual, under "Interfaces".

9.6 Other useful settings (Settings > System Setup)

- Web server: address, port and network name (e.g. rbrowser.local, to open it from the phone without remembering the IP).
- Links (Windows and Linux): nomadnetwork:// and rrc:// links clicked in other programs open in rBrowser.
- Notifications and log: notifications, position of the status bar (top or bottom), sounds, how much it writes in the terminal.
- Internet check: warns you when the internet drops, checking at the interval you choose ("Check every"). Off at first.
- Remote access: opening rBrowser from outside home (tunnel or DDNS: afraid.org/FreeDNS, DuckDNS, dynv6). Switch it on only with the password on.

9.7 Starting with the computer (Settings > System Setup)

"Autostart on boot", off at first: rBrowser starts by itself when the computer starts, no admin rights needed. Windows and macOS at login, Linux at boot (Raspberry Pi too). If you installed the Linux system service (section 3.2) it shows as on here; to remove it: sudo systemctl disable rbrowser. Not on Docker (the container restarts by itself) and not on Termux (it needs the Termux:Boot app). The log of the last start is in settings/autostart.log.

9.8 Guest mode (Settings > Guest mode)

A second password to let others try rBrowser: whoever logs in as a guest browses, but does not change the settings and does not see your keys. Chat and messenger for guests are switched on separately.

10. UPDATING rBROWSER2

10.1 Installed with git clone

With "Check for updates at start" on (Settings > About), when a new version is out an orange "New update!" button appears in the top bar, and a mark on About. There "Update now" downloads the new version and restarts rBrowser. If requirements.txt has changed, it first asks you to run the package install again.

By hand, in the terminal:

```
git pull
pip3 install -r requirements.txt
```

(or run the installer again), then restart rBrowser.

"Files changed by hand" / git pull refuses

You changed some of the program's files: git does not overwrite them. To go back to the original files (your settings, identities and messages are NOT touched, they live in separate folders):

```
git checkout -- .
```

then git pull again.

10.2 Installed from zip

- Download the new zip and extract it into a new folder.
- Copy your things from the old folder to the new one:

```
settings/
identities/
nomadnet_browser_identity
cache/       (saved pages, optional)
downloads/   (downloaded files, optional)
```

- Run the installer again (or pip install -r requirements.txt) and start rBrowser from the new folder.

10.3 Docker

```
git pull
docker compose build rbrowser
docker compose up -d
```

QUICK SUMMARY

```
1. Download:  git clone https://github.com/fr33n0w/rBrowser2.git
              cd rBrowser2
2. Install:   Windows: install.bat
              Linux, Pi, Mac, Termux: bash install.sh
              Docker: bash docker.sh
              By hand: pip3 install -r requirements.txt
              (Debian/Pi OS: --user --break-system-packages, or venv)
3. Start:     python3 rBrowser.py   (Windows: python rBrowser.py,
              venv: .venv/bin/python rBrowser.py)
              In its own window: add --app
4. Open:      http://localhost:5000
5. Settings:  password if the network is not only yours, LXMF name,
              RRC nick, then switch on the clients you want.
6. No nodes?  rnstatus: you need at least one server Up (better 3).
7. Update:    Settings > About > Update now (with git clone).
```

USEFUL LINKS

rBrowser2 on GitHub

https://github.com/fr33n0w/rBrowser2

Python (download)

https://www.python.org/downloads/

Git for Windows

https://git-scm.com/download/win

Termux (F-Droid)

https://f-droid.org/packages/com.termux/

rmap.world (map of the Reticulum network)

https://rmap.world

Official Reticulum manual

https://reticulum.network/manual/

Reticulum Italia on Telegram

https://t.me/reticulumitalia

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